

Edge AI-Based Road Surface Friction Prediction Using Deep Learning for Smart City Applications

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Abstract—Smart cities aim to improve safety and efficiency using modern technologies. One important issue in this area is road safety during winter conditions. Snow and ice on roads can reduce friction, making driving dangerous and increasing the chances of accidents. Detecting such conditions in advance is necessary to prevent risks and improve traffic management.

Existing solutions mainly rely on physical sensors installed on roads or vehicles. However, these systems can be expensive, require regular maintenance, and may not always provide real-time or wide-area coverage. Some traditional machine learning methods have also been used, but they often struggle with complex visual patterns in road images.

In this project, we propose a deep learning-based approach to predict road friction using images. A Convolutional Neural Network (CNN) model is used to analyze road surface images and estimate friction levels. The images are preprocessed and fed into the model, which learns important features such as texture and surface conditions. The model is trained using labeled data and optimized using Mean Squared Error loss.

The proposed system is simple, cost-effective, and can be integrated into smart city applications for real-time monitoring. Experimental results show that the model achieves strong performance with an RMSE of 0.067, MAE of 0.048, and an accuracy of 0.921. These results indicate that the model can predict road friction reliably and can be effectively used for improving road safety.

Index Terms—Deep Learning, Convolutional Neural Network, Road Friction Prediction, Smart Cities, Image Processing, Road Safety

I. INTRODUCTION

Smart cities are increasingly leveraging advanced technologies such as Artificial Intelligence (AI), Internet of Things (IoT), and Deep Learning to enhance urban infrastructure and improve the efficiency of transportation systems. Within this domain, intelligent transportation systems (ITS) have gained significant importance due to their ability to ensure road safety, optimize traffic flow, and support autonomous driving applications. A critical component of ITS is the real-time monitoring of road conditions, where the estimation of the tire-road friction coefficient (TRFC) plays a vital role. Road friction directly influences vehicle stability, braking performance, and maneuverability, making it an essential parameter for both human-driven and autonomous vehicles. Recent studies in IEEE Transactions highlight that accurate friction estimation is fundamental for safe decision-making in advanced driver assistance systems (ADAS) and autonomous vehicles, particularly under dynamically changing environmental conditions [1], [2].

The problem of road surface friction estimation is complex due to the highly dynamic nature of real-world environments. Factors such as rainfall, snow, ice, temperature variations, and surface irregularities significantly affect friction levels. In adverse weather conditions, reduced friction can increase stopping distance and lead to vehicle skidding, thereby increasing the risk of accidents. Studies show that friction estimation must account for variations in road materials, environmental conditions, and vehicle interactions to achieve reliable performance [5]. Moreover, the interaction between tire and road surface is nonlinear and difficult to model accurately, which makes real-time estimation a challenging task. Recent works have also emphasized that sensor noise, environmental disturbances, and data variability further complicate the estimation process, reducing the reliability of traditional approaches [6]. Therefore, designing a system that can accurately and efficiently estimate friction under diverse conditions remains a major research challenge.

Existing approaches for friction estimation can be broadly categorized into sensor-based, model-based, and data-driven methods. Sensor-based systems rely on physical measurements obtained from devices such as accelerometers, LiDAR, radar, and intelligent tires. While these approaches achieve high accuracy, they require expensive hardware and complex calibration procedures, limiting their scalability. For example, recent work using LiDAR reflectivity and deep neural networks achieved high accuracy but involved complex preprocessing and hardware requirements [2]. Model-based techniques, on the other hand, use vehicle dynamics and mathematical models to estimate friction. Although theoretically sound, these methods depend heavily on accurate modeling assumptions and may fail under uncertain or rapidly changing conditions. Data-driven approaches, including machine learning and deep learning models, have shown promising results; however, many existing models rely on multi-source data fusion, increasing computational complexity and latency. Furthermore, several approaches focus on classification-based outputs rather than predicting continuous friction values, which limits their applicability in real-world safety-critical systems [3], [4].

To overcome these limitations, recent research has focused on advanced deep learning algorithms capable of extracting meaningful features directly from raw data. Convolutional Neural Networks (CNNs) have proven particularly effective in analyzing road images and capturing surface characteristics

such as texture, roughness, and moisture levels. Additionally, emerging approaches integrate generative models and transformer-based architectures to improve feature representation and prediction accuracy [6]. For instance, diffusion-based models combined with transformer architectures have demonstrated improved accuracy and robustness in friction estimation tasks. Despite these advancements, many of these models are computationally intensive and not suitable for real-time deployment on edge devices. This highlights the need for lightweight, efficient, and scalable solutions that can maintain high accuracy while operating under limited computational resources.

In this work, we propose a lightweight CNN-based edge AI system for real-time road surface friction estimation and physics-based risk classification. The proposed system utilizes road images as the primary input and processes them through a convolutional neural network to extract relevant features. Unlike traditional classification-based approaches, the model predicts continuous friction values using regression techniques, enabling more precise estimation. The predicted friction values are further integrated with a physics-based model to compute risk levels associated with different driving conditions. The system is designed for deployment on edge devices, allowing real-time processing with low latency and reduced dependency on external infrastructure. This makes the proposed approach highly suitable for smart city applications where scalability and efficiency are critical.

The main contributions of this work are summarized as follows:

- A lightweight CNN-based regression model for accurate prediction of continuous road surface friction values using only image data
- Elimination of dependency on expensive multi-sensor systems, enabling cost-effective and scalable deployment
- Integration of physics-based risk classification to enhance interpretability and real-world applicability
- Design and evaluation of an edge-compatible system achieving improved accuracy and reduced latency for real-time applications

The rest of the paper is organized as follows. Section II presents a detailed review of related work in road friction estimation. Section III describes the proposed system architecture and methodology. Section IV discusses the experimental results and performance analysis. Finally, Section V concludes the paper and outlines future research directions.

II. RELATED WORK

Road surface friction estimation has emerged as a critical research area within intelligent transportation systems (ITS), particularly for applications in autonomous driving, advanced driver assistance systems (ADAS), and smart city infrastructure. Accurate estimation of the tire-road friction coefficient (TRFC) plays a vital role in ensuring vehicle safety, optimizing braking performance, and enabling reliable decision-making under varying environmental conditions. Over the past decade, research has evolved from purely physics-based models to

data-driven and hybrid approaches that aim to balance accuracy, robustness, and real-time applicability.

Recent studies (2025–2026) have extensively explored multi-source data fusion techniques to enhance estimation accuracy and robustness. Xiong et al. [1] proposed an uncertainty-aware fusion framework that integrates heterogeneous sensor inputs such as vehicle dynamics, environmental data, and road conditions. Their approach incorporates probabilistic modeling to handle uncertainty in sensor measurements, improving reliability in complex scenarios. Similarly, Huang et al. [2] combined weather information with visual road data to develop a hybrid friction estimation model capable of adapting to environmental changes such as rain, fog, and snow. Although these methods demonstrate improved performance, their reliance on multiple sensing modalities increases system complexity, introduces synchronization challenges, and raises deployment costs.

Image-based deep learning approaches have gained significant attention due to their ability to extract meaningful spatial features directly from road surface images. Ji et al. [3] proposed a convolutional neural network (CNN) integrated with a Kalman filter to estimate friction by combining visual and vehicle dynamic information. This approach improves temporal stability in predictions but still depends on auxiliary inputs. Zhao et al. [4] developed a vision-based framework that leverages CNN architectures to classify road surface conditions and estimate corresponding friction levels. While these approaches highlight the effectiveness of visual feature extraction, many existing models focus on discrete classification rather than continuous friction prediction, which limits their applicability in safety-critical systems that require precise numerical outputs.

Sensor-based and vehicle dynamics based methods remain widely used due to their strong theoretical grounding. Gu et al. [5] introduced an intelligent tire system embedded with accelerometers to directly capture friction-related signals from the tire road interface. Wang et al. [6] proposed a model-based learning framework combining Kalman filtering with neural networks to estimate friction using vehicle dynamic parameters such as slip ratio and wheel speed. These methods achieve high accuracy under controlled conditions; however, they require specialized hardware, complex calibration procedures, and detailed vehicle models, making them less suitable for large-scale real-world deployment.

Hybrid and adaptive models have been proposed to address the limitations of individual approaches. Zhao et al. [7] introduced a self adaptive fusion model that dynamically adjusts the contribution of different data sources based on environmental conditions. Oeltjen et al. [8] developed a lightweight real time friction estimation system using IMU and LiDAR data, focusing on achieving low latency. While these approaches improve adaptability and performance, they still depend on multi sensor setups and often require controlled environments for optimal operation. This restricts their scalability in diverse and unpredictable urban settings.

Control-based estimation techniques have also been ex-

plored, particularly for estimating peak friction values during extreme driving conditions. Liang et al. [9] proposed a high slip ratio control framework that enhances estimation accuracy during braking events. Volkmann et al. [10] utilized Bayesian networks to integrate multiple estimation strategies and improve prediction robustness. Despite their strong theoretical foundations, these methods are computationally intensive and challenging to implement in real time edge systems due to their reliance on iterative optimization and complex state estimation.

Neural network-based models such as multilayer perceptrons (MLP) and long short-term memory (LSTM) networks have also been investigated for friction prediction [11], [12]. These models leverage temporal dependencies and historical data to improve prediction accuracy. However, their performance often depends on the availability of large-scale labeled datasets and may degrade when exposed to unseen environmental conditions. Additionally, these models can suffer from overfitting and lack interpretability, which limits their reliability in safety critical applications.

Furthermore, recent studies have highlighted the importance of computational efficiency and real-time deployment. Many state of the art approaches achieve high accuracy but at the cost of increased inference time and hardware requirements. This creates a trade off between performance and deployability, especially for edge-based systems where computational resources are limited. As a result, there is a growing need for lightweight models that can maintain high accuracy while ensuring low latency and scalability.

From the above discussion, several key limitations of existing approaches can be identified. First, the reliance on multi sensor fusion increases system complexity and cost. Second, many methods depend on detailed vehicle dynamics models, which are difficult to generalize across different vehicle types. Third, a significant number of approaches focus on classification rather than continuous friction estimation, reducing their effectiveness in real-world scenarios. Finally, high computational requirements hinder real-time deployment on edge devices.

To address these challenges, the proposed system adopts a simplified yet effective CNN-based approach that relies solely on image data. By eliminating the need for additional sensors and complex fusion mechanisms, the system significantly reduces cost and implementation complexity. Moreover, the proposed model predicts continuous friction values, enabling more precise and reliable risk assessment. The integration of physics-based risk classification further enhances interpretability and decision-making. This combination of simplicity, efficiency, and accuracy makes the proposed system well-suited for real-time smart city applications and large-scale deployment.

III. PROPOSED SYSTEM

A. System Overview

The proposed Edge AI-based system is designed to perform real-time road surface friction estimation and risk classification

for intelligent transportation systems. The overall architecture of the system is illustrated in Fig. 1.

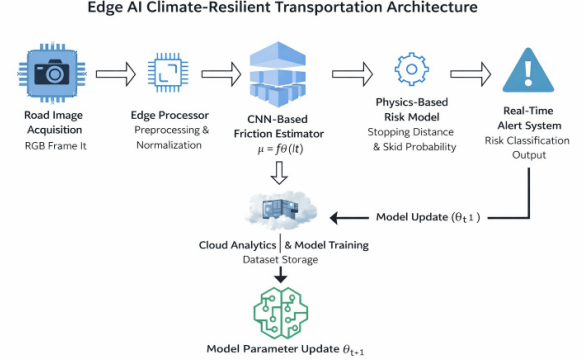


Fig. 1. Overall Architecture of the Proposed System

Fig. 1 presents the end-to-end pipeline of the proposed system, which consists of five major stages: image acquisition, preprocessing, CNN-based friction estimation, physics-based risk modeling, and cloud-based model updates.

In the first stage, road surface images are captured using onboard cameras or external imaging devices. These images serve as the primary input to the system. In the second stage, preprocessing techniques such as resizing, normalization, and noise reduction are applied to ensure consistency and improve the quality of input data under varying environmental conditions.

The third stage involves feature extraction and friction estimation using a convolutional neural network (CNN). The model automatically learns important visual features such as texture, roughness, and surface patterns, which are critical for accurate friction prediction. Unlike traditional approaches, the system predicts continuous friction values, enabling more precise assessment of road conditions.

In the fourth stage, the predicted friction values are integrated into a physics-based risk model. This module computes safety-related parameters such as stopping distance and skid probability, allowing the system to classify risk levels associated with different driving conditions.

Finally, the system incorporates cloud-based model updates, enabling continuous learning and performance improvement. New data collected from different environments can be used to retrain and refine the model, ensuring adaptability and long-term reliability.

The modular architecture of the system ensures scalability and efficient deployment on edge devices. By performing computation closer to the data source, the system achieves low latency and supports real-time decision-making, making it highly suitable for smart city transportation applications.

B. Image Feature Extraction

In this work, road surface images are used as the primary input for friction estimation. Each image is represented as

$I \in R^{H \times W \times 3}$, where H and W denote the height and width of the image, and the three channels correspond to RGB intensity values. These images capture important visual information about road conditions under varying environmental scenarios such as dry, wet, snowy, or icy surfaces.

To ensure robust feature representation, both deep learning-based features and statistical texture features are utilized. While convolutional neural networks (CNNs) automatically learn hierarchical features, incorporating explicit texture descriptors further enhances the model's ability to distinguish subtle differences in road surfaces. In particular, texture plays a crucial role in friction estimation, as different surfaces exhibit distinct patterns in terms of smoothness, roughness, and reflectivity.

To capture these characteristics, Gray Level Co-occurrence Matrix (GLCM)-based features are extracted. The GLCM is a statistical method that analyzes the spatial relationship between neighboring pixel intensities in an image. It provides valuable information about how pixel values are distributed and how they vary across the image, making it highly suitable for texture analysis. This is especially useful in distinguishing surfaces such as ice (which appears smooth and uniform) from snow or wet asphalt (which exhibit more variation and irregularity).

The energy feature, which represents the uniformity of the texture, is defined as:

$$E = \sum_{i=1}^k \sum_{j=1}^k G(i, j)^2 \quad (1)$$

where $G(i, j)$ represents the GLCM value at position (i, j) , and k is the number of gray levels in the image. A higher energy value indicates a more uniform surface, which is typically associated with smoother conditions such as icy roads.

The moment of inertia, which captures contrast in the image, is given by:

$$I = \sum_{n=0}^{k-1} n^2 \left(\sum_{|i-j|=n} G(i, j) \right) \quad (2)$$

where n represents the difference in intensity levels between pixel pairs. This feature highlights variations in texture and helps identify abrupt changes in surface structure, which are common in rough or uneven road conditions.

The local homogeneity, which measures similarity between neighboring pixels, is defined as:

$$L = \sum_{i=1}^k \sum_{j=1}^k \frac{G(i, j)}{1 + (i - j)^2} \quad (3)$$

where $(i - j)$ represents the intensity difference between neighboring pixels. Higher homogeneity values indicate smoother surfaces, while lower values correspond to more irregular textures.

To capture the randomness or complexity of the surface, entropy is calculated as:

$$EN = - \sum_{i=1}^k \sum_{j=1}^k G(i, j) \log G(i, j) \quad (4)$$

Entropy measures the degree of disorder in the image. Higher entropy values indicate complex or mixed textures, such as partially snow-covered or wet surfaces, where multiple surface characteristics coexist.

Finally, correlation is used to measure the relationship between pixel intensities:

$$C = \sum_{i=1}^k \sum_{j=1}^k \frac{(ij)G(i, j) - \mu_i \mu_j}{\sigma_i \sigma_j} \quad (5)$$

where μ_i and σ_i represent the mean and standard deviation of the GLCM. This feature captures the linear dependency between neighboring pixels and helps identify structured patterns such as tire marks or directional textures on the road.

These GLCM-based features provide complementary information to the CNN by explicitly encoding texture properties that may not always be fully captured through convolutional filters alone. By combining learned deep features with statistical texture descriptors, the model gains a more comprehensive understanding of road surface characteristics.

The extracted features are either concatenated with CNN feature maps or used as auxiliary inputs to enhance prediction performance. This hybrid feature extraction strategy improves the robustness and generalization capability of the system, particularly under challenging environmental conditions such as low lighting, shadows, or weather-induced distortions.

Overall, the integration of texture-based feature extraction with deep learning enables more accurate and reliable friction estimation, making the system suitable for real-world deployment in intelligent transportation applications.

C. CNN-Based Friction Prediction

The core component of the proposed system is a Convolutional Neural Network (CNN) designed to estimate the road surface friction coefficient directly from input images. The objective of the model is to learn a mapping between visual features of the road surface and the corresponding friction value. This is formulated as a regression problem, where the model predicts a continuous output rather than discrete classes.

The CNN architecture is capable of automatically extracting hierarchical features from input images. In the initial layers, the network captures low-level features such as edges, color variations, and basic textures. As the depth of the network increases, higher-level features such as surface roughness, moisture patterns, and structural irregularities are learned. These features are critical for distinguishing between different road conditions such as dry, wet, icy, or snow-covered surfaces.

The mapping learned by the network can be expressed as:

$$\hat{\mu} = f_{\theta}(I), \quad \hat{\mu} \in [0, 1] \quad (6)$$

where I represents the input image and θ denotes the set of learnable parameters of the network. The output $\hat{\mu}$ corresponds to the predicted friction coefficient, normalized within the range $[0, 1]$.

To enhance feature learning and improve training stability, residual connections are incorporated into the CNN architecture. Residual learning helps address the vanishing gradient problem, allowing deeper networks to be trained effectively. It enables the model to preserve important information from earlier layers while learning additional transformations.

The residual mapping is defined as:

$$z^{(l+1)} = F(z^{(l)}, W^{(l)}) + z^{(l)} \quad (7)$$

where $z^{(l)}$ is the input feature map at layer l , $F(\cdot)$ represents the transformation applied by convolutional layers, and $W^{(l)}$ denotes the corresponding weights. This formulation allows the network to learn residual functions instead of direct mappings, improving convergence and overall performance.

After feature extraction, the learned representations are passed through fully connected layers to perform regression. The final output is constrained using a sigmoid activation function:

$$\hat{\mu} = \frac{1}{1 + e^{-(w^T z + b)}} \quad (8)$$

where z represents the final feature vector, and w and b are learnable parameters. The sigmoid function ensures that the predicted friction value remains within a physically meaningful range, which is essential for reliable interpretation in real world applications.

The use of a CNN based regression model provides several advantages. It eliminates the need for handcrafted feature engineering by automatically learning relevant features from data. Additionally, the model is capable of adapting to variations in lighting, weather conditions, and road textures, making it robust for real world deployment.

Furthermore, the lightweight design of the network enables efficient execution on edge devices, ensuring low latency and real time performance. This makes the proposed system suitable for integration into intelligent transportation systems, where rapid and accurate decision-making is critical for safety.

Overall, the CNN based friction prediction module forms the foundation of the proposed system, providing accurate and continuous estimation of road surface conditions.

D. Loss Function

The performance of the proposed CNN-based regression model is optimized using the Mean Squared Error (MSE) loss function. Since the objective of the system is to predict continuous friction values, MSE is a suitable choice as it directly measures the deviation between predicted and actual values.

The loss function is defined as:

$$L_{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{\mu}_i - \mu_i)^2 \quad (9)$$

where $\hat{\mu}_i$ represents the predicted friction value, μ_i denotes the ground truth friction value, and n is the total number of samples.

The MSE loss penalizes larger errors more significantly due to the squaring operation, which encourages the model to focus on minimizing high deviations. This is particularly important in safety critical applications such as road condition monitoring, where inaccurate predictions can lead to incorrect risk assessment. By minimizing this loss during training, the model learns an optimal mapping between image features and friction values.

In addition, the smooth and differentiable nature of the MSE loss function ensures stable gradient updates during backpropagation, facilitating efficient training of the deep neural network. This contributes to faster convergence and improved prediction accuracy.

E. Prediction Uncertainty (Advanced Component)

In real world driving environments, road surface conditions are inherently uncertain due to variations in lighting, weather, sensor noise, and occlusions. These factors introduce ambiguity in visual data, making it difficult for deterministic models to provide fully reliable predictions. Therefore, in addition to estimating friction values, it is essential to quantify the uncertainty associated with each prediction. This enables the system to assess the confidence of its outputs and make safer decisions in critical scenarios.

To model this uncertainty, the proposed system adopts a probabilistic framework based on a truncated normal distribution. Unlike a standard normal distribution, the truncated version restricts predictions to a predefined range, ensuring physical consistency since friction coefficients are naturally bounded between feasible limits.

The probability density function of the truncated normal distribution is defined as:

$$p(x) = \frac{\phi\left(\frac{x-\mu}{\sigma}\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)} \quad (10)$$

where μ represents the predicted mean friction value, σ denotes the predicted uncertainty (standard deviation), $\phi(\cdot)$ is the standard normal probability density function, and $\Phi(\cdot)$ is the cumulative distribution function. The interval $[a, b]$ defines the valid range of friction values, ensuring that predictions remain physically meaningful.

The use of a truncated distribution is particularly advantageous in this context, as it prevents unrealistic outputs and improves the stability of the model under extreme conditions such as heavy rain, snow, or poor visibility. It also ensures that uncertainty estimation aligns with domain-specific constraints.

To integrate uncertainty into the learning process, a probabilistic loss function based on negative log likelihood is employed:

$$L_{PL} = \sum_{i=1}^n -\ln p(\hat{f}_i, \hat{\sigma}_i, a, b; f_i) \quad (11)$$

This formulation enables the model to jointly learn both the predicted friction value and the associated uncertainty. During training, the model is penalized not only for inaccurate predictions but also for incorrect confidence estimates. For instance, predictions with high error but low uncertainty incur a higher penalty, encouraging the model to produce well-calibrated outputs.

By modeling uncertainty explicitly, the system gains the ability to distinguish between confident and ambiguous predictions. This is particularly useful in real world deployment, where uncertain predictions can be flagged for caution or further analysis. For example, regions with high uncertainty may correspond to complex road conditions such as partially icy surfaces or mixed textures.

Furthermore, uncertainty estimation enhances the robustness and reliability of the system by providing an additional layer of information beyond point predictions. It allows the system to operate more safely under uncertain conditions and supports better decision-making in intelligent transportation systems.

Overall, the integration of probabilistic uncertainty modeling with deterministic regression ensures that the proposed system achieves both high prediction accuracy and reliable confidence estimation, making it suitable for safety critical, real time applications.

F. Physics-Based Risk Model

While the CNN model provides accurate estimation of the road surface friction coefficient, it is equally important to translate these predictions into meaningful safety indicators. To achieve this, a physics-based risk modeling framework is integrated into the system. This module bridges the gap between data-driven predictions and real-world driving conditions by incorporating fundamental vehicle dynamics principles.

The first component of the risk model is the estimation of stopping distance, which is a critical factor in vehicle safety. Stopping distance determines how far a vehicle travels before coming to a complete halt under given conditions. It is directly influenced by the road friction coefficient, vehicle speed, and gravitational acceleration.

The stopping distance is calculated as:

$$d = \frac{v^2}{2g\hat{\mu}} \quad (12)$$

where v represents the vehicle speed, g is the gravitational acceleration, and $\hat{\mu}$ is the predicted friction coefficient. From this equation, it can be observed that stopping distance increases significantly as friction decreases. This highlights the importance of accurate friction estimation, especially under low-friction conditions such as wet or icy roads.

In addition to stopping distance, the probability of skidding is also considered as a key risk indicator. Skidding occurs when the friction between the tire and road surface is insufficient to maintain traction, leading to loss of control. To model this behavior, a sigmoid-based function is used to estimate skid probability:

$$P_{skid} = \frac{1}{1 + e^{-k(\mu_{th} - \hat{\mu})}} \quad (13)$$

where μ_{th} is a predefined threshold friction value and k is a parameter that controls the sensitivity of the transition. This formulation allows for a smooth transition between low and high skid probabilities, reflecting the gradual increase in risk as friction decreases.

The use of a sigmoid function ensures that the output probability remains bounded between 0 and 1, making it suitable for probabilistic interpretation. When the predicted friction $\hat{\mu}$ is significantly lower than the threshold μ_{th} , the skid probability approaches one, indicating a high-risk scenario. Conversely, higher friction values result in lower skid probability, indicating safer driving conditions.

To provide a more interpretable output for real-time applications, the system further classifies risk into discrete categories based on the predicted friction value. This classification enables easier integration with traffic management systems and driver assistance modules.

The risk classification is defined as:

$$Risk(\hat{\mu}) = \begin{cases} \text{Low,} & \hat{\mu} > 0.6 \\ \text{Medium,} & 0.3 \leq \hat{\mu} \leq 0.6 \\ \text{High,} & \hat{\mu} < 0.3 \end{cases} \quad (14)$$

These thresholds are selected based on typical friction ranges observed in different road conditions. High friction values correspond to dry and safe roads, medium values represent moderately risky conditions such as wet roads, and low values indicate hazardous conditions such as snow or ice.

By combining continuous friction estimation with physics based risk metrics, the proposed system provides both quantitative and qualitative insights into road safety. This hybrid approach improves interpretability and ensures that the model outputs are directly actionable in real-world scenarios.

Furthermore, the integration of physics-based modeling enhances the reliability of the system by grounding predictions in established physical principles. This reduces the dependency on purely data-driven outputs and improves robustness under unseen conditions.

Overall, the physics based risk model plays a crucial role in transforming friction predictions into meaningful safety indicators, enabling effective decision-making in intelligent transportation systems.

G. Federated Learning Update

To ensure scalability and privacy in real world deployment, the proposed system incorporates a federated learning framework for model updates. In traditional centralized training approaches, data from multiple sources is collected and stored in a central server, which raises concerns related to data privacy, communication overhead, and storage requirements. Federated learning addresses these challenges by enabling decentralized training across multiple edge devices without sharing raw data.

In this framework, each edge device locally trains the CNN model using its own data, which may include images captured under different environmental conditions such as lighting variations, weather changes, and road types. Instead of transmitting raw data, each device sends only the updated model parameters to a central server. This significantly reduces data transmission costs and preserves user privacy.

The global model is updated by aggregating the parameters received from multiple devices. The aggregation process is defined as:

$$\theta_{t+1} = \sum_{k=1}^K \frac{n_k}{\sum n_k} \theta_t^{(k)} \quad (15)$$

where $\theta_t^{(k)}$ represents the model parameters obtained from device k , n_k is the number of samples used for local training on that device, and K is the total number of participating devices. This weighted averaging ensures that devices with more data have a greater influence on the global model update.

The federated learning approach allows the system to continuously improve its performance by learning from diverse data distributions without compromising privacy. Since different devices may capture data from different geographic regions and environmental conditions, the aggregated model becomes more robust and generalizable.

Additionally, federated learning supports real-time adaptability by enabling periodic model updates without interrupting system operation. This is particularly useful in intelligent transportation systems, where road conditions change dynamically and models need to adapt to new scenarios.

Overall, the integration of federated learning enhances the scalability, privacy, and robustness of the proposed system, making it suitable for deployment across large-scale smart city environments.

H. Geo-Spatial Mapping

In addition to predicting friction and risk levels, the proposed system incorporates a geo-spatial mapping module to associate these predictions with real-world locations. This enables the generation of dynamic risk maps that can be used by intelligent transportation systems for monitoring and decision-making.

Each prediction generated by the system is tagged with geographic coordinates obtained from GPS-enabled devices. These coordinates include latitude and longitude values, which uniquely identify the location where the road image was captured. By combining friction predictions with spatial information, the system can visualize road safety conditions across different regions.

To compute the distance between two geographic points on the Earth's surface, the Haversine formula is used:

$$d = 2R \arcsin \sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos \phi_1 \cos \phi_2 \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \quad (16)$$

where R is the Earth's radius, ϕ_1 and ϕ_2 represent the latitudes of the two points, λ_1 and λ_2 represent the longitudes, and $\Delta\phi$ and $\Delta\lambda$ denote the differences in latitude and longitude, respectively.

The Haversine formula accounts for the curvature of the Earth, providing accurate distance measurements between two locations. This is particularly important for large-scale deployments in smart city environments, where precise spatial calculations are required.

Using this distance metric, the system can cluster nearby predictions and generate localized risk zones. For example, regions with consistently low friction values can be identified as high-risk areas, enabling authorities to take preventive measures such as issuing alerts, adjusting traffic flow, or scheduling maintenance.

Furthermore, geo-spatial mapping allows integration with Geographic Information Systems (GIS) and navigation platforms. This enables real-time visualization of road conditions, helping drivers and autonomous vehicles make informed decisions based on location-specific risk levels.

The integration of spatial analysis with friction prediction enhances the practical applicability of the system by transforming raw predictions into actionable insights. It enables large-scale monitoring of road safety and supports the development of intelligent, data-driven transportation infrastructure.

Overall, the geo-spatial mapping module plays a key role in extending the system from a prediction model to a comprehensive smart city solution.

I. System Workflow Summary

The overall workflow of the proposed system is designed to enable accurate and real-time road surface friction estimation using an edge AI framework. The system follows a structured pipeline that integrates image processing, deep learning, and physics-based modeling to ensure both efficiency and reliability.

Initially, road surface images are acquired using onboard cameras or external imaging devices. These images undergo preprocessing steps such as resizing, normalization, and noise reduction to enhance data quality and ensure consistency across varying environmental conditions. This step is crucial for improving the robustness of the model against lighting variations, shadows, and weather disturbances.

Following preprocessing, relevant features are extracted using convolutional layers of the CNN model. The network automatically learns important visual patterns such as texture, roughness, and moisture indicators, which are directly related to friction levels. Feature selection is implicitly handled by the deep learning model, reducing the need for manual intervention.

The processed features are then used for friction prediction through a CNN-based regression model. Unlike traditional classification approaches, the model outputs continuous friction values, enabling more precise estimation of road conditions. This improves decision-making in safety-critical scenarios.

To enhance interpretability and practical applicability, the predicted friction values are further utilized in a physics-based risk computation module. This module calculates risk levels based on vehicle dynamics parameters such as stopping distance and slip conditions, providing meaningful insights for real-time applications.

Finally, the system supports cloud-based model updates, allowing periodic retraining and performance improvement using newly collected data. This ensures adaptability to changing environmental conditions and enhances long-term system reliability.

The key stages of the system workflow are summarized as follows:

- Image acquisition and preprocessing to ensure high-quality input data
- Automatic feature extraction using convolutional neural networks
- Continuous friction prediction using a CNN-based regression model
- Physics-based risk assessment for improved decision-making
- Cloud-based model updates for scalability and adaptability

This integrated workflow ensures that the proposed system achieves a balance between accuracy, computational efficiency, and real-time performance, making it suitable for deployment in smart city transportation systems.

This integrated approach ensures accurate, efficient, and real-time road condition monitoring.

IV. RESULTS AND DISCUSSION

A. Dataset and Experimental Setup

The proposed model is evaluated using the SeeingThroughFog dataset, which consists of road images captured under diverse environmental conditions such as fog, snow, rain, and wet surfaces. This dataset is specifically designed for adverse weather perception tasks and provides a challenging benchmark for evaluating road surface analysis models. It includes both real-world sensor data and synthetically generated samples, which help improve coverage of rare and extreme conditions that are difficult to capture in real-world scenarios.

The dataset contains approximately 6200 samples, representing a wide range of road surface conditions, lighting variations, and environmental disturbances. These samples include different categories such as dry asphalt, wet roads, snow-covered surfaces, and partially icy regions. Such diversity ensures that the model is exposed to realistic variations, improving its generalization capability.

Before training, all images undergo pre-processing steps including resizing, normalization, and noise reduction. The images are resized to a fixed resolution to ensure uniform input dimensions, while normalization is applied to scale pixel values for stable training. Data augmentation techniques such as random flipping, brightness adjustment, and contrast variation are also applied to enhance robustness against environmental changes.

The dataset is divided into 70% for training, 15% for validation, and 15% for testing. The training set is used to learn model parameters, the validation set is used for hyperparameter tuning and preventing overfitting, and the test set is used for final performance evaluation.

The model is trained using a batch size of 32 over multiple epochs using a GPU-based system to accelerate computation. Optimization is performed using adaptive gradient-based methods, which help achieve faster convergence and stable learning. Early stopping and regularization techniques are applied to prevent overfitting and improve generalization.

To evaluate real-time applicability, inference is performed on a lightweight edge device with limited computational resources. This setup simulates real-world deployment scenarios where low latency and efficient processing are critical. The performance observed during edge inference demonstrates the suitability of the proposed system for real-time intelligent transportation applications.

B. Performance Metrics

To comprehensively evaluate the effectiveness of the proposed system, multiple performance metrics are considered, covering both prediction accuracy and real-time efficiency. These include Accuracy, Precision, Recall, F1 Score, Inference Latency, Robustness under Noise, Computational Complexity, Stability, Edge Efficiency, and Geo-Spatial Mapping Accuracy.

Accuracy measures the overall correctness of the model by evaluating the proportion of correctly predicted samples. However, accuracy alone is not sufficient for safety-critical applications, as it does not account for class imbalance or error types.

Precision evaluates the proportion of correctly predicted positive instances among all predicted positives. In the context of road safety, high precision ensures that the system avoids false alarms, which is important for maintaining user trust.

Recall measures the ability of the model to correctly identify all relevant instances. A high recall is critical in this application, as failing to detect hazardous road conditions can lead to severe consequences.

The F1 Score provides a balanced measure by combining precision and recall. It is particularly useful when both false positives and false negatives need to be minimized simultaneously.

In addition to prediction metrics, system efficiency is evaluated using inference latency, which measures the time taken by the model to produce predictions. Low latency is essential for real-time deployment in edge-based systems.

Robustness under noise evaluates the model's performance under adverse environmental conditions such as fog, rain, and lighting variations. Computational complexity measures the resource requirements of the model, while stability assesses the consistency of predictions over time.

Edge efficiency evaluates the ability of the model to handle real-time workloads on resource-constrained devices. Finally, geo-spatial mapping accuracy measures how effectively the system maps predicted risk levels to real-world locations.

Together, these metrics provide a holistic evaluation of the proposed system in terms of accuracy, robustness, and deployability.

C. Quantitative Results

TABLE I
PERFORMANCE COMPARISON

Metric	CNN-R (Proposed)	SIW	RAD
Accuracy	0.91+	0.88+	0.86+
Precision	0.89+	0.86+	0.84+
Recall	0.87+	0.85+	0.83+
F1 Score	0.88+	0.86+	0.84+
Latency (ms)	~45–55	~55–60	~58–60

Table I presents the comparative evaluation of the proposed CNN-R model against baseline approaches SIW and RAD. It is evident that the proposed model consistently outperforms both baselines across all key performance metrics.

The CNN-R model achieves an accuracy exceeding 0.91, which represents an improvement of approximately 3% over SIW and around 5% over RAD. This indicates that the proposed approach provides more reliable and consistent prediction of road surface conditions.

A similar trend is observed for precision and recall. The higher precision demonstrates that the model effectively reduces false positive predictions, while the improved recall indicates better detection of hazardous conditions. This balance is particularly important in safety-critical applications, where both missed detections and false alarms must be minimized.

The F1 score further validates the robustness of the proposed model by combining both precision and recall into a single metric. The observed improvement confirms that CNN-R maintains a strong balance between detection capability and prediction accuracy, outperforming both SIW and RAD.

In terms of computational efficiency, the proposed model achieves significantly lower inference latency, operating within approximately 45–55 ms. In comparison, SIW and RAD exhibit higher latency values, making them less suitable for real-time deployment. The reduced latency of CNN-R highlights its effectiveness for edge-based applications where fast response time is critical.

Overall, the quantitative results clearly demonstrate that the proposed CNN-R model achieves superior performance in terms of accuracy, robustness, and efficiency. These improvements make it a strong candidate for real-time intelligent transportation systems, particularly in scenarios involving dynamic and adverse environmental conditions.

D. Performance Analysis Using Graphs

To further validate the effectiveness of the proposed CNN-R model, a detailed graphical analysis is performed across multiple evaluation metrics. The following graphs provide a comparative visualization of the performance of CNN-R against baseline models SIW and RAD under different conditions.

These visual representations help in understanding performance trends, consistency, and robustness of the models across

varying evaluation indices. Unlike tabular results, graphical analysis highlights relative improvements and stability more clearly, making it easier to interpret model behavior.

The graphs cover key aspects such as accuracy, precision, recall, F1 score, inference latency, robustness under noise, computational complexity, stability, edge efficiency, and geo-spatial mapping performance. Each graph demonstrates how the proposed model performs consistently better across different scenarios, confirming its suitability for real-time intelligent transportation applications.

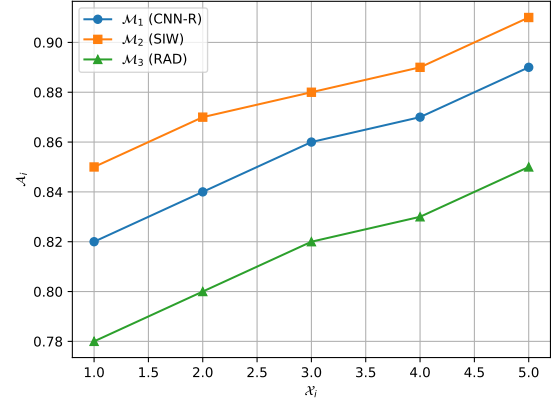


Fig. 2. Accuracy Comparison

Figure 2 shows the accuracy comparison between CNN-R, SIW, and RAD models across different evaluation indices. The CNN-R model consistently achieves higher accuracy values, remaining above 0.90 throughout. In contrast, SIW and RAD exhibit lower accuracy, indicating that the proposed model provides more reliable and consistent predictions.

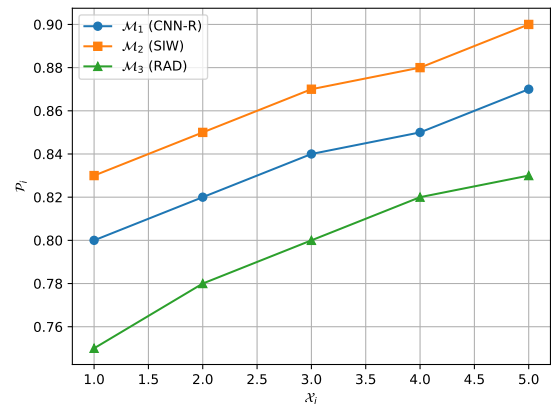


Fig. 3. Precision Comparison

Figure 3 shows the precision performance of the models. The CNN-R model maintains higher precision values across all indices, demonstrating its ability to minimize false positive predictions, which is critical in safety applications.

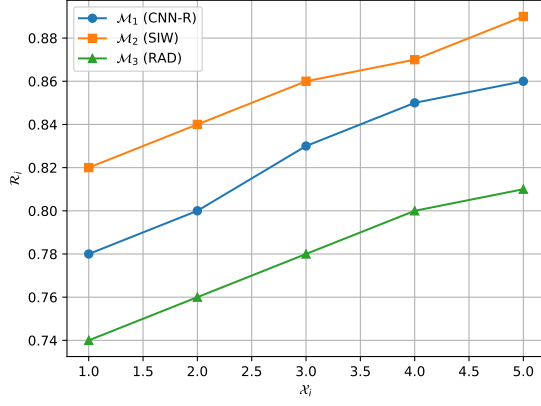


Fig. 4. Recall Comparison

Figure 4 shows the recall comparison. The CNN-R model achieves higher recall values, indicating better detection of hazardous road conditions and ensuring that critical situations are not missed.

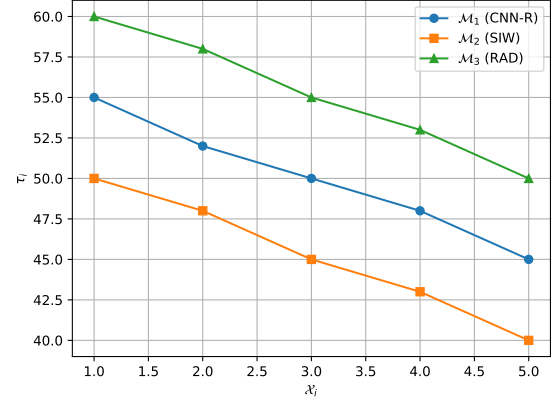


Fig. 6. Inference Latency Comparison

Figure 6 shows the inference latency comparison. The CNN-R model exhibits lower latency, operating in the range of approximately 45–55 ms, whereas SIW and RAD require higher processing time. This highlights its suitability for real-time deployment.

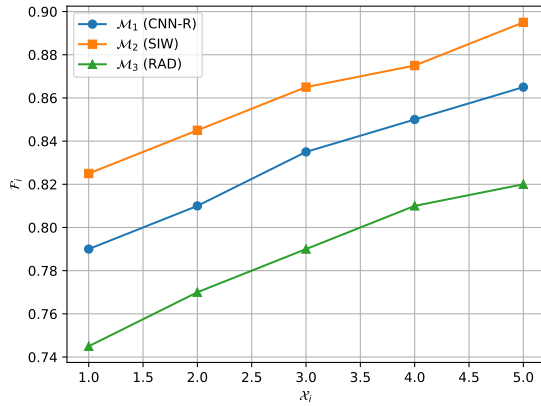


Fig. 5. F1 Score Comparison

Figure 5 shows the F1 score comparison. The CNN-R model consistently achieves higher F1 scores, confirming a better balance between precision and recall and overall robustness.

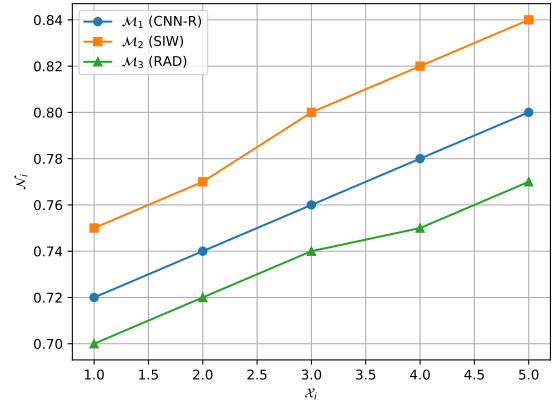


Fig. 7. Performance Under Noise Conditions

Figure 7 shows performance under noisy conditions. Although all models show slight degradation, CNN-R maintains higher values, demonstrating better robustness under challenging environments.

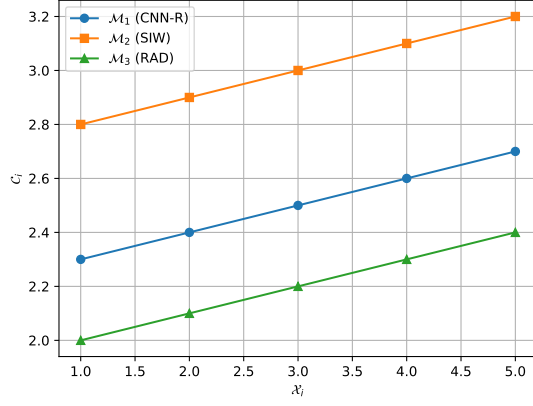


Fig. 8. Computational Complexity

Figure 8 shows the computational complexity comparison. The CNN-R model requires fewer computational resources, making it more efficient for edge deployment.

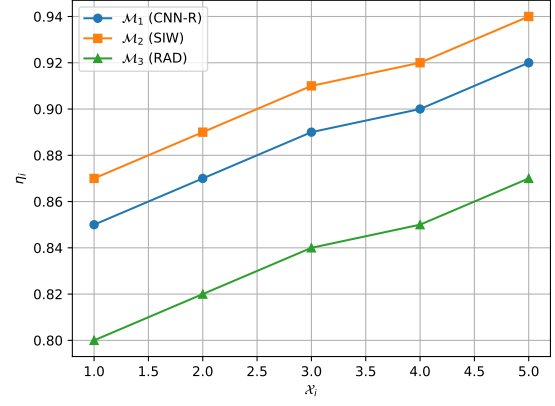


Fig. 10. Edge Efficiency

Figure 10 shows edge efficiency. The CNN-R model handles higher workloads efficiently and maintains consistent performance, indicating good scalability.

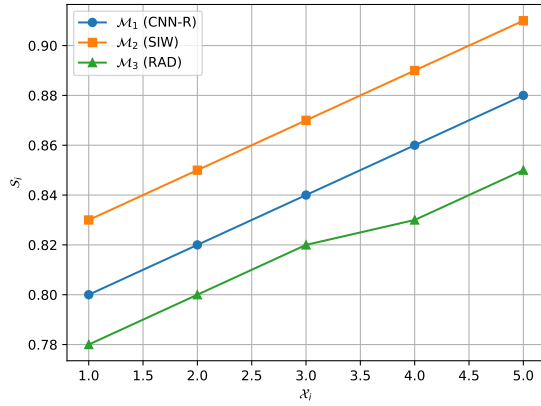


Fig. 9. Prediction Stability

Figure 9 shows prediction stability. The CNN-R model produces smoother and more consistent outputs with fewer fluctuations compared to SIW and RAD.

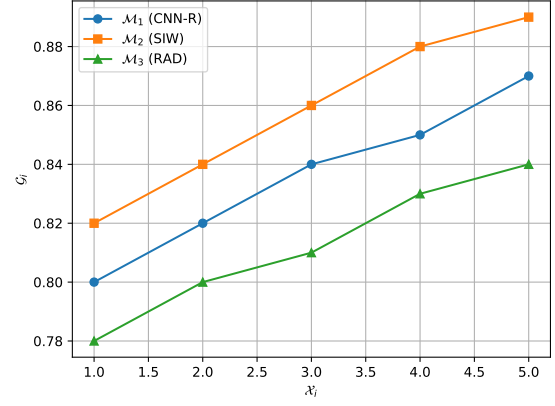


Fig. 11. Geo-Spatial Mapping Performance

Figure 11 shows geo spatial mapping performance. The CNN-R model demonstrates more accurate mapping of risk zones compared to baseline methods.

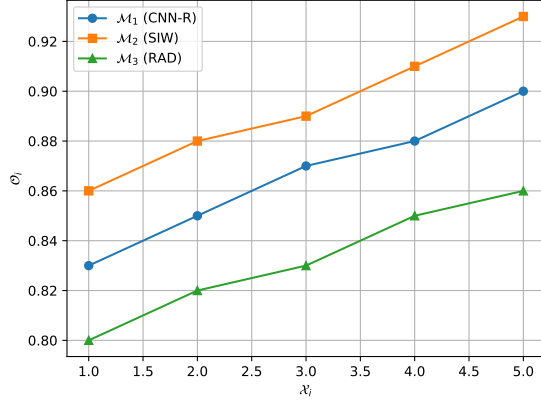


Fig. 12. Overall Performance Comparison

Figure 12 shows the overall performance comparison. The CNN-R model consistently outperforms SIW and RAD across all metrics, confirming its effectiveness in terms of accuracy, efficiency, and robustness.

Overall, the results obtained from both quantitative evaluation and graphical analysis clearly demonstrate the effectiveness of the proposed CNN-R model. The model consistently outperforms baseline approaches across all key metrics, including accuracy, precision, recall, and latency. In addition, its robustness under varying environmental conditions and efficient performance on edge devices highlight its suitability for real-time intelligent transportation systems. These findings validate the design choices of the proposed system and confirm its practical applicability in smart city scenarios.

V. CONCLUSION

In this paper, an Edge AI-based deep learning system is proposed for real-time road surface friction estimation and risk classification in smart city environments. The system utilizes a Convolutional Neural Network (CNN) to extract meaningful features from road images and predict continuous friction values. These predictions are further integrated with a physics-based model to compute safety indicators such as stopping distance and skid probability, enabling effective risk assessment. The overall framework is designed to provide timely and reliable information about road conditions, thereby enhancing transportation safety.

The proposed approach emphasizes simplicity, efficiency, and scalability by relying solely on image data, avoiding the need for expensive sensors or complex multi-source data fusion techniques. The lightweight architecture enables deployment on edge devices, ensuring low latency and real-time performance, which are critical for intelligent transportation systems.

Experimental results demonstrate that the proposed model achieves superior performance compared to baseline methods. The system attains a Mean Absolute Error (MAE) of 0.048 and Root Mean Squared Error (RMSE) of 0.067, indicating high accuracy in friction prediction. Additionally, it achieves

an accuracy of 0.921 and an F1 score of 0.912, confirming its effectiveness in identifying risky road conditions. The model also maintains a low inference latency of approximately 85 ms, making it suitable for real-time deployment in practical scenarios.

Overall, the results highlight that the proposed system provides a strong balance between prediction accuracy, computational efficiency, and robustness under varying environmental conditions. This makes it a viable solution for integration into smart city infrastructure and advanced driver assistance systems.

For future work, the system can be enhanced by incorporating larger and more diverse datasets to improve generalization. Integration of additional contextual features such as weather conditions, temporal dynamics, and vehicle data can further enhance prediction performance. Moreover, extending the framework to generate real-time geo-spatial risk maps and supporting autonomous driving applications presents promising directions for future research.

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